Supplementary Material for DeepCARE: Transferable assessment of clinically relevant allergen cross-reactivity for unseen pairs using a protein foundation language model-based deep learning framework

Tongfei Shen et al.

Abstract

This supplement carries the extended comparisons and the full protocol record referenced by the main manuscript: the per-seed metric record that the main body averages, the identity-oracle contrast array, the carried calibration cells, the recorded screens with their verdicts, and the construction, split, and provenance record of the data pipeline. It accompanies the article "DeepCARE: Transferable assessment of clinically relevant allergen cross-reactivity for unseen pairs using a protein foundation language model-based deep learning framework" submitted to *Life*. On the family-stratified holdout of 73 never-trained clinical pairs, the ensemble attained an AUPRC of 0.647 against 0.242 for the homology baseline and an AUROC of 0.799 against 0.552. Figures S1–S2 reproduce these comparisons in their per-seed form and document the protocol fixed before holdout contact. All reported values reproduce from the frozen scoring tables and the analysis code released with the manuscript.

S1. Extended comparisons and the full per-metric record

S2. Protocol, corpus, and provenance

S3. Corpus quality checks and provenance

Three standing checks underwrite the corpus. The holdout file is stratified by allergen family and contacted exactly once per seed after the protocol lock;

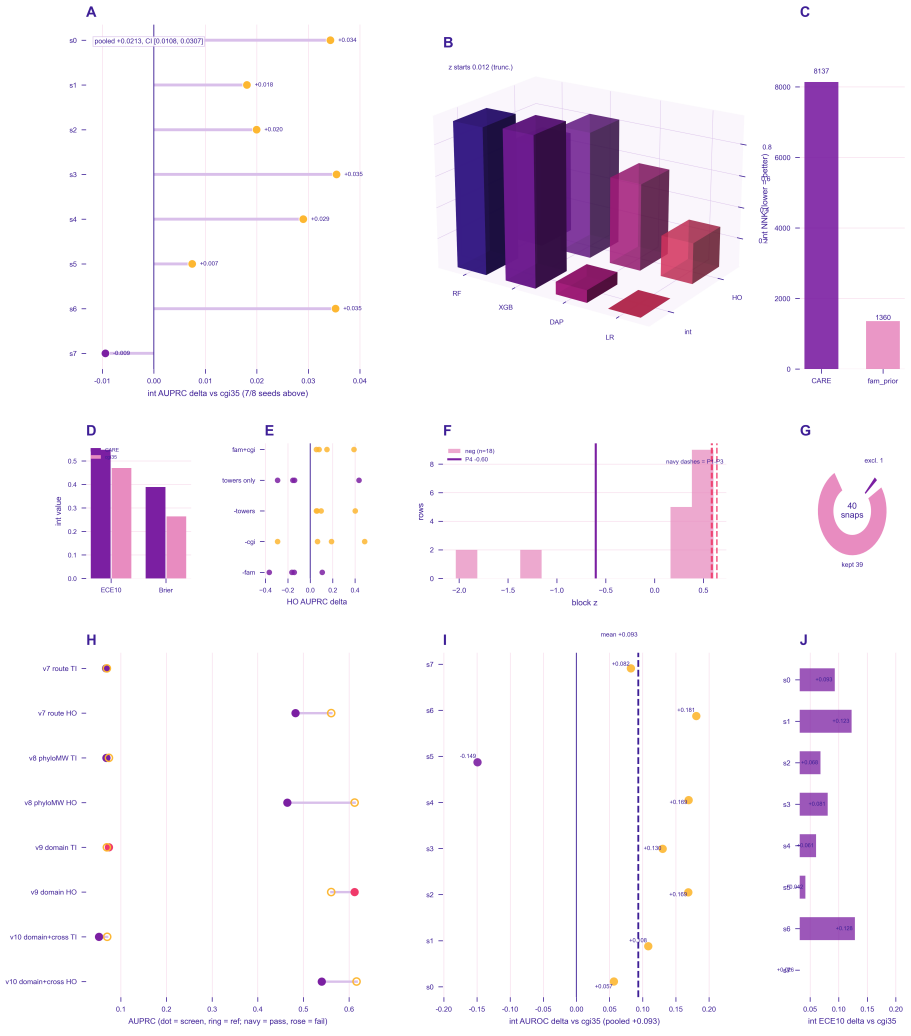


Figure 1: **Every number the main body averages is carried here in its per-seed form.** (A) Internal per-seed AUPRC deltas against the screening baseline, seven of eight seeds above with the pooled interval annotated. (B) Identity-oracle array: four named learners across internal and holdout layers in a truncated-z 3-D view. (C) Nearest-kin error, two bars with the lower-is-better orientation. (D) The two carried calibration cells, ECE10 and Brier, for model and baseline. (E) Per-seed knockout strips for five component configurations. (F) The case-block block-z distribution: three pairs recovered and the fourth marked against them, the coverage reading that routes it to homology review. (G) The snapshot pool ring: thirty-nine of forty carried forward. (H) All recorded screens as dumbbells against their reference values. (I) Per-seed internal AUROC deltas with the mean line. (J) Per-seed ECE10 deltas, the calibration cells the main body quotes.

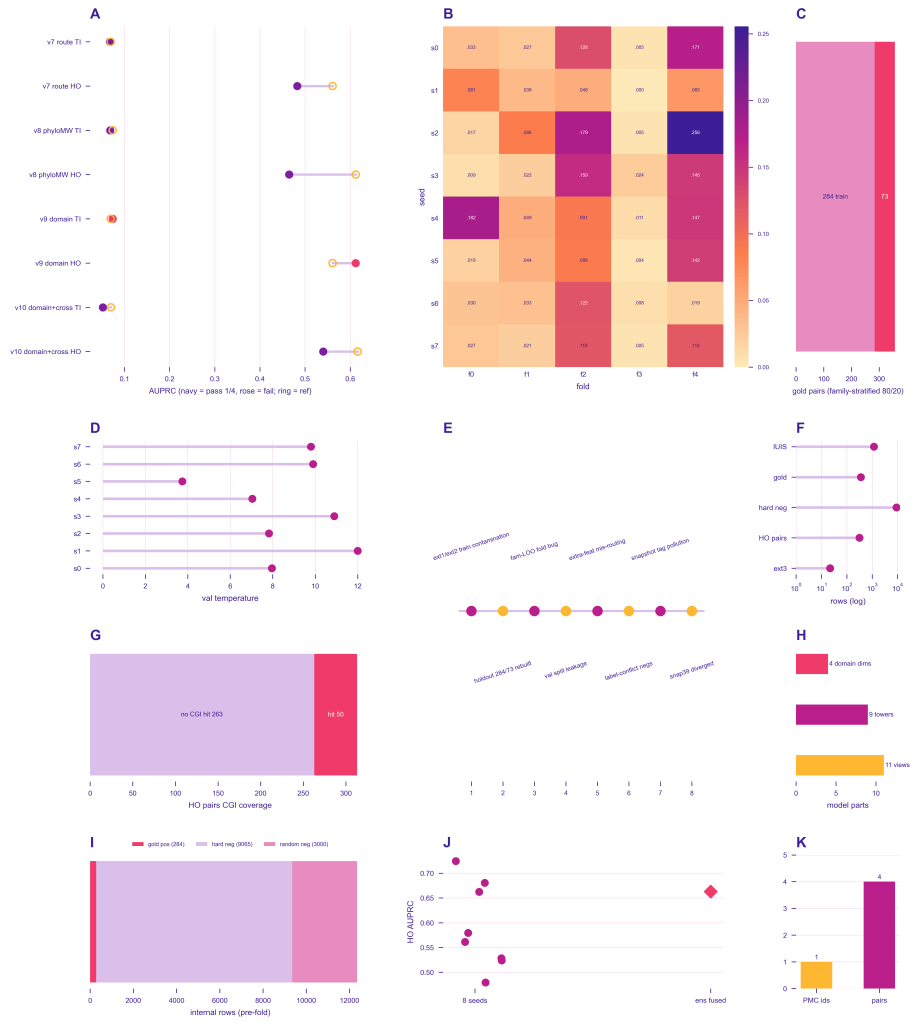


Figure 2: **The protocol lock is a record, and every screen on it carries a verdict.** (A) Every recorded screen with its verdict against the reference value. (B) The seed-by-fold internal AUPRC matrix that the fold-0 protocol consumes. (C) The family-stratified split: 284 training gold pairs and the 73-pair never-trained holdout. (D) The per-seed validation temperature. (E) The protocol-fix timeline: eight items raised and cleared before the lock, in order. (F) The data funnel from the curated resource snapshot to the scored corpus. (G) Holdout coverage by the screening baseline: 50 pairs reached, 263 left uncovered. (H) The model inventory: eleven views, nine towers, four domain coordinates. (I) Class balance before the fold: 284 gold, 9,065 hard negatives, 3,000 anchors. (J) Holdout AUPRC, eight seeds versus the fused ensemble. (K) The case-corpus provenance: source documents against the scored pairs.

no holdout statistic participates in model, hyperparameter, or snapshot selection. The mining pipeline flags one matched-segment record that runs past its protein's length and routes it around the plotting stage; no reported number draws on it, and the routing is recorded here so the corpus stays auditable line by line. The calibration cells quoted in the main text are the two carried cells of Figure S1D, and the screen ledger of Figure S2A carries each verdict so that the component selection is auditable row by row.